PROJECT DEMO

Subject : Generative AI

Group : A2 (11:45)

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Project 11 : Mental Health First-Response Chatbot

1.Team Members:

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2. Problem Statement:

"University counseling services often have a delay of 2–4 weeks, leaving students without immediate mental health support. This project aims to provide an AI-based first-response chatbot that offers instant emotional support, basic coping strategies, and crisis detection."

3. Objective:

- *Provide instant support*
- *Detect emotional state*
- *Identify crisis situations*
- *Suggest coping techniques*
- *Escalate emergencies*

4. Scope:

"This system is a first-response triage and support tool and is NOT a replacement for professional medical diagnosis or therapy."

5. Features:

- *Chat-based interface*
- *Sentiment detection (BERT)*
- *Crisis detection system*
- *CBT-based suggestions*
- *Mood tracking dashboard*
- *Emergency escalation (SMS)*

6. System Architecture:

Flow:

```
User Input
   ↓
BERT Sentiment + Crisis Detection
   ↓
LangChain Memory + RAG
   ↓
Response / Twilio Alert
```

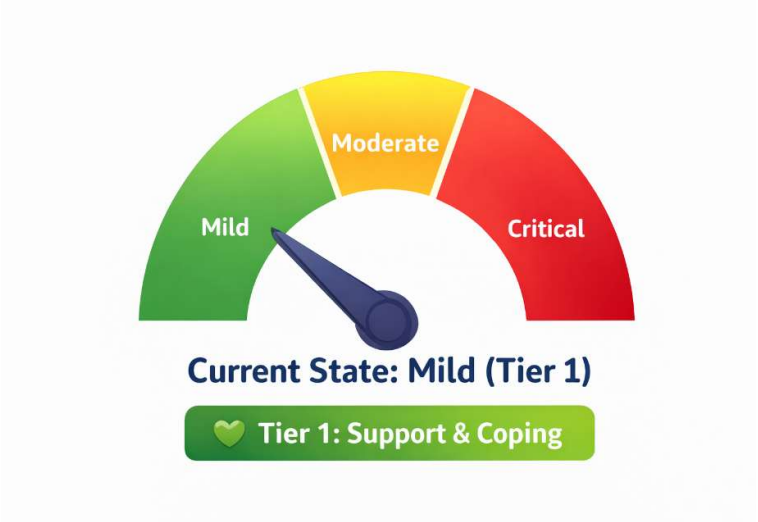
```
User: "I want to end my life"
   ↓
Crisis Detection (Tier 3)
   ↓
Twilio API triggered
   ↓
SMS sent to emergency contact
```

- *Input → analyzed → processed → response generated*

7. Crisis Escalation Tiers:

- ***Tier 1:(Green)***
- *Mild stress*
- *Normal chatbot support*
- ***Tier 2:(Yellow)***
- *High distress*
- *Stronger suggestions*
- ***Tier 3:(Red)***
- *Self-harm signals*
- *Emergency response + SMS*

Demo images(generated using AI):



8. Technology Stack:

- *Frontend → React / Flutter*
- *Backend → FastAPI*
- *AI → BERT + LangChain*
- *Database → PostgreSQL / MongoDB*
- *Vector DB → FAISS / ChromaDB*
- *API → Twilio*

9. Database Design:

PostgreSQL:

- *Users table*
- *Sessions table*
- *Mood logs table*

10. Team Roles:

- ***NLP & Sentiment Analysis:***
  Deepsikha Patra, Mehek Mahapatra
  (Handles BERT model, sentiment analysis, and crisis detection)
- ***Backend Development:***
  ISTA RANJAN BARAL, AYUSHMAN PANDA
  (Develops FastAPI server, database, and APIs)
- ***Frontend Development:***
  Lochan Kumar Nayak, Subham Satua
  (Builds chat UI and mood dashboard)
- ***Integration & Testing:***
  Chaganti Kavya, Ayush Ranjan
  (Integrates modules, handles Twilio alerts, testing)
- ***Documentation & Presentation:***
  Debasish Panigrahy
  (Prepares report, PPT, and manages demo)

11. Project Workflow:

Step-by-step working:

1. *User enters message*
2. *Sentiment analyzed*
3. *Crisis checked*
4. *Response generated*
5. *Data stored*

12. Ethical Considerations:

- *No diagnosis*
- *Data privacy*
- *Emergency override*
- *User safety priority*

13. Conclusion:

"This project aims to bridge the gap between emotional distress and professional help by providing immediate, safe, and intelligent support."